

test. Electromagnetic interference (EMI) can disturb device readings or even render the device dysfunctional. Thus EMC testers are in great demand for electromedical testing.

Sumit Sharma, marketing manager, Good Will Instrument Co., mentions, "Earlier, T&M manufacturers used to provide only spectrum analysers for EMC testing. Now, the trend is to offer a package of pre-compliance EMI testing and EMC evaluation together. It includes all the necessary accessories and probes to analyse whether the device is compliant to set EMC regulations, and, if there is any abnormality, to identify the source of abnormality."

Madhukar Tripathi, sales manager, Anritsu India, cites an example from their T&M roster: "Anritsu offers EMI test setups using spectrum analyser MS2712E or MS2830A for medical electronics devices. MS2830A signal analyser comes with an EMI pre-compliance test feature."

Wireless chip testing

With wireless connectivity trending in medical devices, T&M equipment are required to evaluate connectivity modules on chips. Separate equipment are available for Wi-Fi modules, Bluetooth chips and GSM chips.

Test sets capable of analysing Bluetooth modules of medical devices are in demand. Tripathi mentions, "We have upgraded our Bluetooth test set to provide better evaluation. The device supports T&M for basic rate, enhanced data rate and Bluetooth Low Energy (BLE) measurements. It analyses transmission power, frequency, modulation and receiver sensitivity.



Anritsu's MS2830A signal analyser

The equipment have improved to the level where a single button can initiate a measurement script. Also, these can test a medical device in less than 10 seconds."

Current flow and pressure testing

Insulation is an important safety aspect in medical electronics. Thus it becomes an essential test parameter. A spokesperson from Tektronix mentions, "Measuring leakage current of medical electronics devices is very important since these devices are often in contact with the human body. So, leakage current testers are greatly utilised by medical instrument testers."

Saamil S. Patel of Spectrum Med-tech adds, "Measuring the pressure of medical devices is very important for optimal maintenance. We use specific pressure-measuring instruments. Device pressure can be evaluated in two parts: low pressure and high pressure. For medical devices used in angiography and angioplasty, the low

pressure should be around 450 pounds per square inch (psi) and the high pressure reading should stay within 1200psi."

More upgrades

Wireless connectivity and cloud connectivity too are on the rise in T&M instruments. Amitesh says, "Test equipment becoming Internet-connected. Devices like spectrum analysers and EMC testers can be controlled via smartphones from anywhere—the next room or even the next building. Moreover, results are stored on a cloud server. So, this data can be accessed by authorised personnel from anywhere."

Furthermore, addition of touch-screen and single-button control is enabling ease of use. Automated testing and upgraded software ensure hassle-free, error-free and reliable T&M.

Selection tips

Just like medical devices, T&M tools should also comply with safety standards such as International Electrotechnical Commission (IEC) authorised ISO 60601 series. Prospective buyers should ensure that their product of choice is certified. Also, in order to maintain a smooth technological and asset transition, ensure that the device is compatible with your existing systems. Additionally, assess service availability options and rate contract offered by the manufacturer. It is advisable to purchase T&M devices from renowned brands to ensure reliability and after-sales service. **EFY**



The ProSim 8 ECG device tester from Fluke